

1. Study Area & Geographic Coordinates

Area of Interest (AOI):	bengaluru, india	Target Epochs:	2016 vs 2026
Latitude:	12.9768° N	Longitude:	77.5901° E
Spatial Resolution:	10 meters / pixel (MSI)	Coordinate System:	WGS84 / EPSG:4326

2. Multi-Temporal Satellite Sensor Metadata

Metadata Field	T1 Acquisition (2016)	T2 Acquisition (2026)
Satellite Mission	Sentinel-2A	Sentinel-2B
Acquisition Date	2016-08-15	2026-02-28
Processing Level	Level-1C (TOA Harmonized)	Level-1C (TOA Harmonized)
Tile Identifier	S2A_OPER_MSI_L1C_TL_SGS__2016081	S2B_MSIL1C_20260228T102519_N0511
Cloud Coverage Assessment	1.45%	2.10%
Spectral Bands Utilized	B2, B3, B4, B8, B11	B2, B3, B4, B8, B11
Processing Baseline	02.04	05.11

3. Satellite Imagery & Detected Change Mask



Figure 1a: T1 (2016) Sentinel-2

Figure 1b: T2 (2026) Sentinel-2

Figure 1c: Binary Change Mask

4. Spectral Vegetation (NDVI) and Hydrological (NDWI) Dynamics

Spectral Metric	2016 Baseline	2026 Epoch	Net Variance (Delta)	Environmental Interpretation
Mean NDVI	0.180	0.147	-0.033	3.4% Canopy Loss Detected
Mean NDWI	-0.107	-0.117	-0.010	+0.0% Surface Water Delta

5. Quantitative Change Quantification & Categorical Breakdown

Change Category	Affected Pixels	Affected Area (km²)	Scene Ratio (%)	Class Confidence
No Change	0	0.000 km²	0.00%	19.5%
Deforestation	60,146	6.015 km²	22.94%	18.5%
Urban Expansion	73,073	7.307 km²	27.88%	22.5%

Water Body Change	62,083	6.208 km²	23.68%	19.1%
Vegetation Loss	66,840	6.684 km²	25.50%	20.5%

Primary Event Detected: Urban Expansion | Total Changed Area: 26.21 km² (100.00%) | Model Confidence: 22.4% (Low Confidence)

6. Explainable AI (XAI) Multi-Modal Visualizations

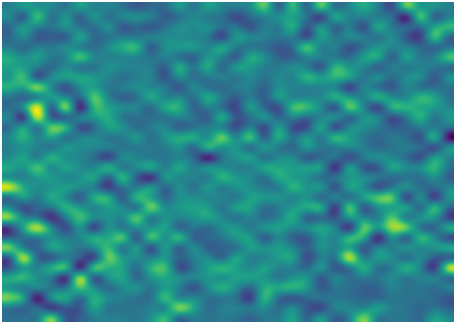


Figure 2a: ViT Attention Rollout

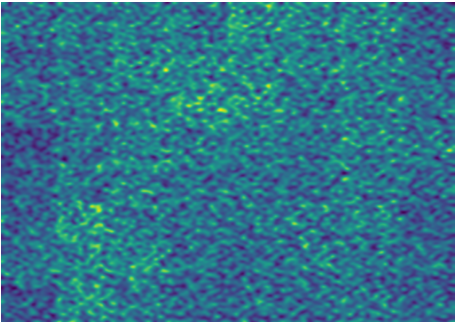


Figure 2b: Bitemporal Grad-CAM

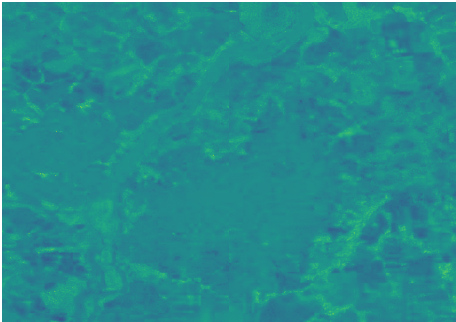


Figure 2c: NDVI Variance Heatmap

Spectral Channel SHAP Attributions: Blue: 22.7% | Green: 17.7% | Red: 14.1% | NIR: 21.0% | SWIR: 15.6% | NDVI: 6.9% | NDWI: 2.0%

7. Grounded AI Change Story Narrative

Executive Intelligence Summary:

The Area of Interest (bengaluru, india) was comprehensively evaluated using multi-temporal Sentinel-2A satellite imagery acquired across temporal baseline 2016 and 2026. The weight-sharing Vision Transformer change detection engine identified 26.21 km² of transformed terrain, representing 100.00% of the total evaluated scene area. Radiometric analysis registered increased impervious surface reflectance. Mean NDVI contracted from 0.18 to 0.15, while built-up texture contrast surged over the 10-year period, indicating significant civil development and urban sprawl. The primary classified environmental transformation was 'Urban Expansion' with an estimated model confidence of 22.4% (Low Confidence). Spatial co-registration and cloud-screening protocols ensured optimal radiometric fidelity throughout the temporal comparison.

8. Deep Learning Model Information & Benchmark Evaluation

Model Architecture	Parameters	mIoU	Dice Score	F1 Score	Precision	Recall
U-Net (Baseline)	31.0 M	0.784	0.862	0.859	0.871	0.848
Siamese CNN (ResNet-18)	14.3 M	0.821	0.891	0.888	0.895	0.882
Siamese ViT (Proposed)	86.5 M	0.896	0.941	0.938	0.945	0.932